

Dr. Thejus Mahajan

CONTACT INFORMATION

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<https://scholar.google.com/citations?user=PJkZwAwAAAAJ>

RESEARCH PROFILE Ocean-biogeochemical modeller with experience developing plankton dynamics models within the GOTM-FABM/ERGOM framework. Research focus on implementing biological parameterizations—including trait diffusion, individual-based population dynamics, and life-cycle processes—in coupled physical-biogeochemical models. Experienced in running climate scenario simulations (hindcast and IPCC projections) and analyzing large-scale model output. Experienced in contributing to regional-scale modelling of marine ecosystem responses under present-day and projected climate change conditions.

WORK EXPERIENCE

- **University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences**

Post-doctoral Scientist – Marine Ecosystem Modelling
August 2021 – January 2025

- Developed the novel “Cyanobacteria Life Cycle” (CLC) model within the ERGOM/GOTM-FABM framework (Fortran). Implemented a trait-diffusion scheme (adapted from Beckmann 2019) for phytoplankton thermal adaptation.
- Built both Eulerian and Lagrangian (Individual-Based Model) approaches for phytoplankton population dynamics, incorporating stochastic processes and trait evolution.
- Conducted hindcast and forecast simulations using ERA5 data and projected climate scenarios (IPCC) to assess ecosystem responses.
- Executed high-performance computing (HPC) workflows on the SLURM cluster and analyzed large NetCDF dataset ensembles using Python (NumPy, xarray).
- Translated a Fortran continuous model into a high-performance Python engine using JAX, enabling massive GPU/TPU parallelization via vectorized, branchless operations.
- Parental leave: 01/2024–12/2024.

- **German Language Studies**

Goethe B1 Certification
February 2025 – April 2025

- **Helmholtz-Zentrum Hereon, Geesthacht, Germany**

Guest Scientist
May 2025 – October 2025

- Continued research on cyanobacteria life-cycle modelling in collaboration with the Ecosystem Modelling Department (Prof. Dr. Kai Wirtz), finalizing the modelling work for publication.

- **CQ Beratung + Bildung, Berlin, Germany**

Further Training – Bioinformatics and Biostatistics
August 2025 – February 2026

- Acquired additional computational and statistical methods (Python, R, biostatistics) complementing existing modelling skill set.
- Capstone: Built a reproducible Nextflow (DSL2) pipeline for Hepatitis Delta Virus sequence analysis — automated NCBI download, MAFFT alignment, and trimAl trimming with Conda-managed dependencies (GitHub).

- **HealthTwist GmbH, Berlin, Germany**

Internship – Large-Scale Data Analysis & Pipeline Engineering
February 2026 – April 2026

- Refactored a monolithic 1,834-line R data pipeline (tidyverse) for Germany’s interventional radiology quality registry (DeGIR); 143,000+ patient records from ~300 clinics. Reduced to 1,348 lines (–26.5%) with byte-identical output verified via `identical()`.
 - Externalised 257 hard-coded correction rules into 8 configuration CSV files, enabling non-programmer editing of business rules.
 - Replaced 357 deprecated function calls across 4 pipeline files, modernising the codebase to current tidyverse standards.
 - Created the `degirtools` R package (9 exported functions, roxygen2 docs, `devtools::check()` passing) by extracting reusable logic from a 767-line helper file.
 - Discovered 2 pre-existing bugs through systematic code analysis; documented both and preserved original behaviour for supervisor review — demonstrating production-code discipline.
 - Built an interactive R Shiny dashboard (6 modules: overview, interventions, complications, radiation doses, success rates, about) using plotly, DT, and bslib. Deployed to shinyapps.io with GDPR-safe synthetic data.
 - Wrote Tidymodels teaching materials (Quarto/GitHub) replacing the legacy caret framework for ML workflows in R.
- **Tutorwaves Solutions Inc / Freelance, Kerala, India**
Physics Educator October 2018 – March 2021
 - **Université Paris-Saclay, Institut des Sciences Moléculaires d’Orsay, France**
Ph.D. Researcher October 2015 – September 2018
 - Led atomic collision experiments; processed terabytes of data and modeled fragmentation using C++.

TEACHING
EXPERIENCE

- **Marine Ecosystem Modelling** (BSc level)
University of Hamburg, 2022–2023
Regular teaching contribution.
- **“From 0D to 1D” – Advanced Marine Ecosystem Modelling** (MSc level)
University of Hamburg, 2023
2-day guest lecture as part of “Introduction to Biological Oceanography and Fisheries Science” course.

EDUCATION

- **Ph.D. in Astrochemistry** (October 2015 – September 2018)
Institut des Sciences Moléculaires (ISMO), Université Paris-Saclay, France
- **M.Sc. in Physics** (July 2012 – December 2014)
National Institute of Technology Calicut, India
First Class with Distinction
- **B.Sc. in Physics** (June 2009 – April 2012)
University of Calicut, India

TECHNICAL SKILLS

Modelling works:	Frame-	GOTM-FABM, ERGOM, Individual-Based Models (IBM), trait-diffusion models
Programming:		Fortran, Python (JAX, NumPy, Pandas, xarray), R (tidyverse, R Shiny, devtools/roxygen2), C++, BASH
HPC & Infra:		SLURM, MPI, Hardware Acceleration (TPU/GPU via JAX), Linux/Unix, Git
Data & Analysis:		NetCDF, HDF5, ERA5 reanalysis data, large-scale model output analysis, climate scenario analysis

PUBLICATIONS

Google Scholar: <https://scholar.google.com/citations?hl=en&user=PJkZwAwAAAAJ>

In preparation:

- T. Mahajan, E. Schaum, K. Wirtz. *Cyanobacteria life-cycle modelling and individual-based model trait-diffusion results*. In preparation.

Peer-reviewed (5 articles):

1. T. Mahajan, et al. *Journal of Physics: Conference Series*, 1412:142026, 2020.
2. T. Idbarkach, et al. *Astronomy & Astrophysics*, 628:A75, 2019.
3. T. Mahajan, et al. *Journal of Physics B*, 2019.
4. T. Idbarkach, et al. *Journal of Physics B*, 51(24):245201, 2018.
5. T. Idbarkach, et al. *Journal of Physics: Conference Series*, 1412(11):112028, 2020.

LANGUAGES

English (proficient), German (B1 – Goethe Certified), French (intermediate), Malayalam (native)

REFERENCES

- **Prof. Dr. Elisa Schaum**

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Relation: Postdoctoral collaborator

- **Prof. Dr. Kai Wirtz**

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Relation: Guest Scientist supervisor